



Series EF1GH/C



SET~4

रोल नं.							
Roll No.							

प्रश्न-पत्र कोड
Q.P. Code

465

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

व्यावहारिक गणित APPLIED MATHEMATICS

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निर्धारित समय : 3 घण्टे

Time allowed : 3 hours

अधिकतम अंक : 80

Maximum Marks : 80

नोट / NOTE :

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 23 हैं।
Please check that this question paper contains 23 printed pages.
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में 38 प्रश्न हैं।
Please check that this question paper contains 38 questions.
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book before attempting it.
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) *This question paper contains **38** questions. **All** questions are **compulsory**.*
- (ii) *This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*
- (iii) *In **Section A**, Questions no. **1** to **18** are multiple choice questions (MCQs) and questions number **19** and **20** are Assertion-Reason based questions of **1** mark each.*
- (iv) *In **Section B**, Questions no. **21** to **25** are very short answer (VSA) type questions, carrying **2** marks each.*
- (v) *In **Section C**, Questions no. **26** to **31** are short answer (SA) type questions, carrying **3** marks each.*
- (vi) *In **Section D**, Questions no. **32** to **35** are long answer (LA) type questions carrying **5** marks each.*
- (vii) *In **Section E**, Questions no. **36** to **38** are case study based questions carrying **4** marks each.*
- (viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 2 questions in Section D and 2 questions in Section E.*
- (ix) *Use of calculators is **not** allowed.*

SECTION A

This section comprises multiple choice questions (MCQs) of 1 mark each.

- 1. If $100 \equiv x \pmod{7}$, then the least positive value of x is :
 - (a) 6
 - (b) 4
 - (c) 3
 - (d) 2
- 2. In a kilometre race, A beats B by 50 metres or 10 seconds. The time taken by A to complete the race is :
 - (a) 90 seconds
 - (b) 120 seconds
 - (c) 190 seconds
 - (d) 200 seconds



3. If a man rows 32 km downstream and 14 km upstream in 6 hours each, then the speed of the stream is :
- (a) 2 km/h (b) 1.5 km/h
- (c) 2.5 km/h (d) 2.25 km/h
4. If $-3x + 17 < -13$, then :
- (a) $x \in (10, \infty)$ (b) $x \in [10, \infty)$
- (c) $x \in (-\infty, 10]$ (d) $x \in [-10, 10]$
5. If A and B are two matrices such that $AB = A$ and $BA = B$, then B^2 is equal to :
- (a) B (b) A
- (c) I (d) O
6. If $A = \begin{bmatrix} 5 & x \\ y & 0 \end{bmatrix}$ is a symmetric matrix, then :
- (a) $x = 0, y = 5$ (b) $x = 5, y = 0$
- (c) $x = y$ (d) $x + y = 0$
7. The value of x for which the points (2, -1), (-3, 4) and (x, 5) are collinear, is :
- (a) -4 (b) 4
- (c) 2 (d) -2



8. If $x + y = 8$, then the maximum value of xy is :
- (a) 12 (b) 16
(c) 20 (d) 24
9. The function $f(x) = x^x$, $x > 0$ is decreasing in the interval :
- (a) $(-\infty, e)$ (b) $(0, e)$
(c) $\left(0, \frac{1}{e}\right)$ (d) $\left[\frac{1}{e}, \infty\right)$
10. $\int \frac{1}{x + x \log x} dx$ is equal to :
- (a) $1 + \log x + C$ (b) $x + \log x + C$
(c) $x \log (1 + \log x) + C$ (d) $\log (1 + \log x) + C$
11. If the supply function is $p = 4 + x$, then the producer's surplus when 12 units are sold, is :
- (a) 72 (b) 64
(c) 76 (d) 46
12. The order of the differential equation corresponding to family of curves $y = Ae^{3x} + Be^{-3x}$ is :
- (a) 1 (b) 2
(c) 3 (d) 4



13. If 'm' is the mean of Poisson distribution, then $P(r = 0)$ is given by :
- (a) e^{-m} (b) e^m
(c) e (d) m^{-e}
14. Normal distribution is symmetric about :
- (a) variance (b) standard deviation
(c) mean (d) covariance
15. For a student's t-test, the test statistic t is given by :
- (a) $t = \bar{x} - \mu$ (b) $t = \frac{\bar{x}}{s}$
(c) $t = \frac{\bar{x} - \mu}{s}$ (d) $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}}$
16. For the given values 15, 23, 28, 36, 41, 46, the 3-yearly moving averages are :
- (a) 24, 29, 35, 41 (b) 22, 28, 35, 41
(c) 22, 29, 35, 41 (d) 24, 28, 35, 41
17. The effective rate of return which is equivalent to nominal rate of 8% p.a. compounded quarterly is : [Given $(1.02)^4 = 1.0824$]
- (a) 8.16% (b) 7.95%
(c) 8.24% (d) 8.5%
18. The maximum value of the function $z = 7x + 5y$, subject to constraints $x \leq 3, y \leq 2, x \geq 0, y \geq 0$, is :
- (a) 21 (b) 10
(c) 31 (d) 37



Questions number **19** and **20** are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
- (c) Assertion (A) is true but Reason (R) is false.
- (d) Assertion (A) is false but Reason (R) is true.

19. Assertion (A) : The slope of the normal to the curve $y = 2x^2 - 5x$ at $x = -1$ is -1 .

Reason (R) : The slope of the normal to the curve $y = f(x)$ at point (α, β) is given by $(x - \alpha) + \left(\frac{dy}{dx}\right)_{(\alpha, \beta)} \cdot (y - \beta) = 0$.

20. Assertion (A) : If A is a square matrix of order 3 such that $|\text{adj } A| = 144$, then the value of $|A|$ is ± 12 .

Reason (R) : If A is an invertible matrix of order n, then $|\text{adj } A| = |A|^{n-1}$.

SECTION B

This section comprises very short answer (VSA) type questions of 2 marks each.

21. (a) Solve the inequality :

$$\frac{3}{5}x - \frac{2x - 1}{3} > 1, x \in W$$

OR

(b) Solve the inequality :

$$-\frac{2}{3} < -\frac{x}{3} + 1 \leq \frac{2}{3}, x \in R$$





22. Write the matrix $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ as a sum of a symmetric and a skew symmetric matrix.

23. (a) It is given that 2% of screws manufactured by a company are defective. Using Poisson distribution, find the probability that a packet of 100 screws contains no defective screw.

(Given : $e^{-2} = 0.14$)

OR

(b) If the standard deviation of a Poisson variable X is $\sqrt{3}$, then find $P(X > 0)$. [Use : $e^{-3} = 0.05$]

24. Find the present value of a sequence of payments of ₹ 1000 made at the end of every 6 months and continuing forever, if money is worth 8% per annum compounded semi-annually.

25. The value of a machine purchased two years ago, depreciates at the annual rate of 10%. If its present value is ₹ 97,200, find.

(i) its value after 3 years;

(ii) its value when it was purchased.

SECTION C

This section comprises short answer (SA) type questions of 3 marks each.

26. Formulate the following problem as an LPP :

A small firm manufactures gold rings and chains. The total number of rings and chains manufactured per day is at most 24. It takes 1 hour to make a ring and 30 minutes to make a chain. The maximum number of hours available per day is 16. If the profit on a ring is ₹ 300 and that on a chain is ₹ 190, find the number of rings and chains that should be manufactured per day, so as to earn the maximum profit.



27. A person can row a boat at 5 km/h in still water. It takes him thrice as long to row upstream as to row downstream. Find the rate at which the stream is flowing.
28. A container has 50 litres of juice in it. 5 litres of juice is taken out and is replaced by 5 litres of water. This process is repeated four more times. What is the amount of juice left in the container after final replacement ?
[Take $(0.9)^5 = 0.59049$]

OR

In a 1000-metre race, A, B and C get Gold, Silver and Bronze medals respectively. If A beats B by 100 metres and B beats C by 100 metres, then by how many metres does A beat C ?

29. A fair coin is tossed 9 times. Find the probability of getting.
- (i) exactly 5 tails;
 - (ii) at least 5 tails;
 - (iii) at most 5 tails.

OR

Let X denote the number of hours a person watches T.V. during a randomly selected day. The probability that X can take the values x_i , has the following form, where k is some unknown constant.

$$P(X = x_i) = \begin{cases} 0.2, & \text{if } x_i = 0 \\ kx_i, & \text{if } x_i = 1 \text{ or } 2 \\ k(5 - x_i), & \text{if } x_i = 3 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Find the value of k .
- (ii) Find : $P(X = 2)$, $P(X \geq 2)$ and $P(X \leq 2)$.



30. 2000 students appeared in an examination. Distribution of marks is assumed to be normal with mean 30 and standard deviation 6.25. How many students are expected to get marks

- (i) between 20 and 40 ?
- (ii) less than 25 ?

[Use : $P(0 \leq z \leq 1.6) = 0.4452$, $P(0 \leq z \leq 0.8) = 0.2881$]

31. The mean of IQs of a group of 18 students from one area of a city was found to be 95 with a standard deviation of 10, while the mean of IQs of a group of 12 students from another area of the city was found to be 100 with a standard deviation of 8. Test whether there is a significant difference between the IQs of two groups of students at 1% level of significance. [Use : $t_{28}(0.01) = 2.763$]

SECTION D

This section comprises long answer (LA) type questions of 5 marks each.

32. (a) Determine for what values of x , the function $f(x) = x^4 - \frac{x^3}{3}$ is strictly increasing or strictly decreasing.

OR

- (b) A firm has the following total cost and demand functions :

$$C(x) = \frac{x^3}{3} - 7x^2 + 111x + 50 \text{ and } x = 100 - p$$

- (i) Find the total revenue function in terms of x .
- (ii) Find the total profit function P in terms of x .
- (iii) Find the profit maximizing level of output of x .
- (iv) What is the maximum profit, taking rupee as a unit of money ?



33. (a) A company establishes a sinking fund to provide for the payment of ₹ 1,00,000 debt, maturing in 4 years. Contributions to the fund are to be made at the end of year. Find the amount of each annual deposit if interest is 18% per annum. [Use $(1.18)^4 = 1.9388$]

OR

- (b) A firm bought a machinery for ₹ 7,40,000 on 1st April, 2020 and ₹ 60,000 is spent on its installation. Its useful life is estimated to be of 5 years. Its scrap value at the end of 5 years is estimated to be ₹ 40,000. Find the amount of annual depreciation and the rate of depreciation.
34. A person takes a housing loan worth ₹ 10,00,000 at an interest rate of 6% p.a compounded monthly. He decided to repay the loan by equal monthly instalments in 15 years. Calculate the EMI, using
- (i) flat rate method,
- (ii) reducing balance method.
- [Given : $(1.005)^{-180} = 0.4074824$]
35. A library has to accommodate two different types of books on a shelf. The books are each 6 cm and 4 cm thick and each weighs 1 kg and $1\frac{1}{2}$ kg respectively. The shelf is 96 cm long and can support a weight of atmost 21 kg. How should the shelf be filled with the books of two types in order to include the greatest number of books ? Formulate it as an L.P.P and so solve it graphically.



SECTION E

This section comprises 3 case study based questions of 4 marks each.

Case Study – 1

- 36.** 10 students were selected from a school on the basis of values for giving awards and were divided into three groups. The first group comprises hard workers, the second group has honest and law abiding students, and the third group contains vigilant and obedient students. Double the number of students of the first group added to the number in the second group gives 13, while the combined strength of the first and the second group is four times that of the third group. Assume that x , y and z denote the number of students in first, second and third group respectively.

Based on the above information, answer the following questions :

- (a) Write the system of linear equations that can be formulated from the above described situation. 1
- (b) Write the coefficient matrix, say A . 1
- (c) (i) Write the matrix of cofactors of every element of matrix A . 2

OR

- (c) (ii) Determine the number of students of each group. 2

Case Study – 2

- 37.** A company notes that higher sales of a particular item, which it produced, is achieved by lowering the price charged. As a result, the total revenue from the sales at first rises as the number of units sold increases, reaches the highest point, and then falls off. The pattern of revenue is described by the relation $y = 40,00,000 - (x - 2000)^2$, where y is the total revenue and x is the number of units sold.

Based on the above information, answer the following questions :

- (a) Find what number of units sold maximizes total revenue. 2
- (b) What is the amount of this maximum revenue ? 1
- (c) What would be the total revenue if 2500 units were sold ? 1



Case Study – 3

38. The following data shows the percentage of rural, urban and sub-urban Indians who have high speed internet connection at home.

Year	Rural	Urban	Sub-urban
2016	3	9	9
2017	6	18	17
2018	9	21	23
2019	16	29	29
2020	24	38	40

Based on the above information, answer the following questions :

- (a) Derive straight-line trend by the method of least squares for the rural students. 2

OR

- (a) Derive straight-line trend by the method of least squares for the urban Indians. 2
- (b) What is the forecast for the year 2021 for urban group using trend equation ? 1
- (c) What is the forecast for the year 2021 for rural group using trend equation ? 1